

CSE 464 Project Part 1: Graph Parser

Author: Sai Mahesh Nomula (snomula)

GitHub Repository: <https://github.com/Sai-2811/CSE464-2026-smahesh>

Hi! This is my submission for Part 1 of the CSE 464 course project. For this assignment, I built a Java application that can parse, manipulate, and export directed graphs using DOT files.

My Environment Setup

To get everything working, I used the following setup on my machine:

- **Java:** JDK 11 (or higher)
- **Build Tool:** Maven 3.8+
- **IDE:** IntelliJ IDEA Community Edition
- **Graphviz:** Installed system-wide so I could use the `dot` command line tool for rendering images directly from my Java code.

Building and Running My Code

If you want to build my project from scratch and run the tests, you can use Maven:

```
mvn clean package
```

To run the actual application and see it parse the sample graph, add nodes/edges, and generate the final outputs, you can run:

```
mvn exec:java -Dexec.mainClass="edu.asu.cse464.Main"
```

How It Works (Features & Outputs)

I started by creating a sample DOT file (`input.dot`) to test my parser. Here is what my input file looks like:

```
digraph G {  
    a -> b;  
    b -> c;  
    c -> d;  
    d -> a;  
    a -> e;  
    e -> f;  
    e -> g;  
    f -> h;  
    g -> h;  
}
```

Below, I've included snapshots of my terminal output to show that all four features are working properly. To make it easy to read, I've boxed my terminal outputs below.

Feature 1: Parsing the DOT Graph

When my program reads `input.dot` , it successfully identifies all the nodes and directed edges. Here is the snapshot of my terminal printing the parsed graph details:

```
=====
=== Parsed Graph ===
Number of nodes: 8
Node labels: a, b, c, d, e, f, g, h
Number of edges: 9
Edges: a -> b, b -> c, c -> d, d -> a, a -> e, e -> f, e -> g, f -> h, g -> h
=====
```

Features 2 & 3: Adding Nodes and Edges

Next, I implemented the ability to add new nodes and directed edges. I also added validation to make sure duplicate nodes or edges throw an `IllegalArgumentException` .

In my demo, I added a new node `x` , a list of nodes `[y, z]` , and then connected them with edges `x -> y` and `y -> z` . Here is the snapshot of the updated graph printed to my console:

```
=====
=== Updated Graph ===
Number of nodes: 11
Node labels: a, b, c, d, e, f, g, h, x, y, z
Number of edges: 11
Edges: a -> b, b -> c, c -> d, d -> a, a -> e, e -> f, e -> g, f -> h, g -> h, x -> y, y -> z
=====
```

Running the Unit Tests

I wrote JUnit 5 tests to cover all of the graph features (parsing, adding nodes, adding edges, duplicates, and exporting). When I run `mvn test` , all 8 tests pass successfully. Here is the snapshot from my Maven test run:

```
=====
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.asu.cse464.GraphTest
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.055 s -- in edu.asu.cse464.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
=====
```

Feature 4: Exporting Output to DOT and PNG

Finally, my program successfully exports the in-memory graph back out to a `.dot` file and uses Graphviz to render it into a PNG image.

Here is the confirmation from my console: ``text

DOT and PNG exported successfully.

And here is the actual `output.png` image that my program generated:

```
![Output Graph PNG](output/output.png)
```

My GitHub Commits

I made sure to commit my work frequently and logically, keeping each feature in a separate commit as requested by the assignment. You can find the links to my specific commits here:

- **Feature 1 (Parse Graph):** [View Commit](https://github.com/Sai-2811/CSE464-2026-smahesh/commit/ddf4e16)
- **Feature 2 (Add Nodes & Duplicates Check):** [View Commit](https://github.com/Sai-2811/CSE464-2026-smahesh/commit/8307f4c)
- **Feature 3 (Add Directed Edges):** [View Commit](https://github.com/Sai-2811/CSE464-2026-smahesh/commit/373a9b5)
- **Feature 4 (Export to DOT & PNG):** [View Commit](https://github.com/Sai-2811/CSE464-2026-smahesh/commit/42a10c6)
- **Unit Tests:** [View Commit](https://github.com/Sai-2811/CSE464-2026-smahesh/commit/c1f9066)

Thanks for taking the time to review my project! Let me know if you have any questions.